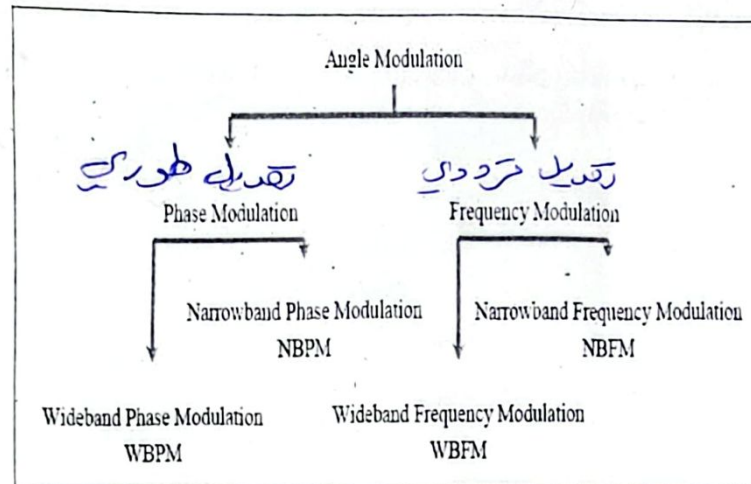


PART 2

Analog Communication - Angle Modulation



The other type of modulation in continuous-wave modulation is **Angle Modulation**. Angle Modulation is the process in which the frequency or the phase of the carrier signal varies according to the message signal.

The standard equation of the angle modulated wave is

$$s(t) = A_c \cos \theta_i(t)$$

Where,

المعدلة لاهي
سوية معدلة
زاوية
زاوية الموجة الحاملة ← A_c ← سعة الموجة الحاملة

A_c is the amplitude of the modulated wave, which is the same as the amplitude of the carrier signal

$\theta_i(t)$ is the angle of the modulated wave

Angle modulation is further divided into frequency modulation and phase modulation.

● **Frequency Modulation** is the process of varying the frequency of the carrier signal linearly with the message signal.

● **Phase Modulation** is the process of varying the phase of the carrier signal linearly with the message signal.

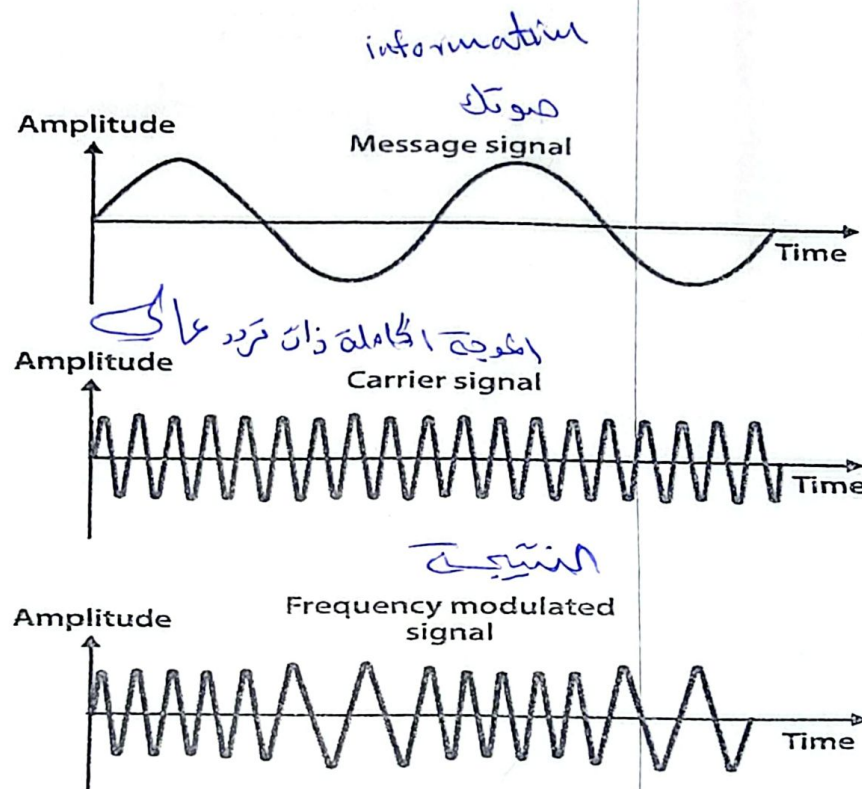
تجعل تردد الموجة الكاملة يتغير بشكل مباشر مع تغير الإشارة المحلومح

تجعل طور الموجة الكاملة يتغير بشكل مباشر مع الإشارة المحلومح

1. Frequency Modulation

In amplitude modulation, the amplitude of the carrier signal varies. Whereas, in **Frequency Modulation (FM)**, the frequency of the carrier signal varies in accordance with the instantaneous amplitude of the modulating signal.

Hence, in frequency modulation, the amplitude and the phase of the carrier signal remains constant. This can be better understood by observing the following figures.



The frequency of the modulated wave increases, when the amplitude of the modulating or message signal increases. Similarly, the frequency of the modulated wave decreases, when the amplitude of the modulating signal decreases. Note that, the frequency of the modulated wave remains constant.

دقیقتاً کدری زهری قوتاً یثیر الإشارة المملوون
على التردد

Mathematical Expression of FM modulation "time domain expression":

(a) Instantaneous frequency $[f_i(t)]$:-

The Instantaneous frequency of FM-signal is mathematically defined as, $f_i(t) = f_c + K_f m(t)$ as shown in figure 1

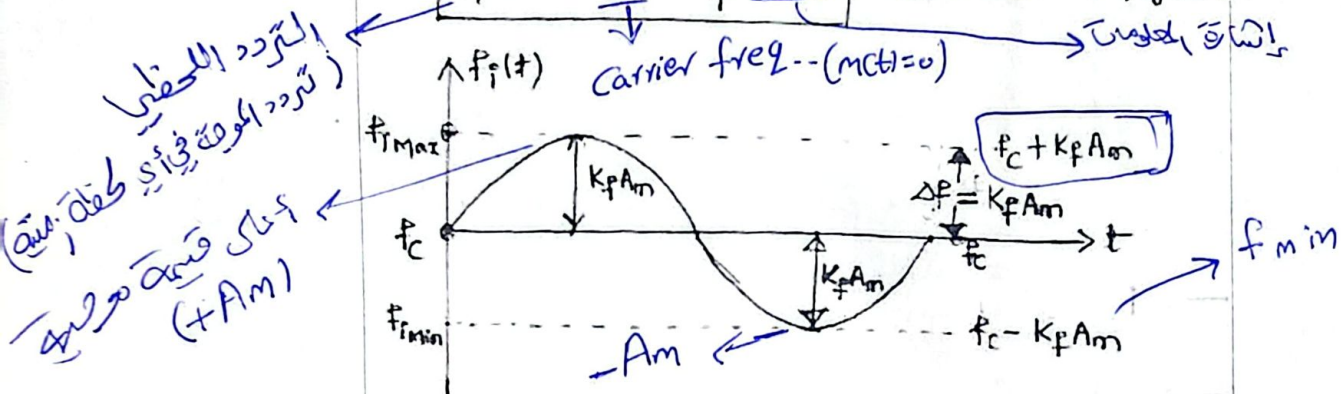


Figure 1: Instantaneous frequency $f_i(t)$ of FM signal for sinusoidal message signal $m(t)$

From figure 1, following observations can be made

→ When $m(t)=0$: $f_i(t) = f_c$ → Same as that of unmodulated carrier

→ At peak value of $m(t)$ i.e., $|m(t)| = A_m$ } $\Rightarrow f_{i(t)_{max}} = f_c + K_f A_m$

∴ Maximum frequency of FM signal is $f_{i(t)_{max}} = f_c + K_f A_m$

maxim plus dipher

The equation for instantaneous frequency f_i in FM modulation is

$$f_i = f_c + k_f m(t)$$

Where,

f_c is the carrier frequency

k_f is the frequency sensitivity

$m(t)$ is the message signal

We know the relationship between angular frequency ω_i and angle $\theta_i(t)$ as

$$\omega_i = \frac{d\theta_i(t)}{dt}$$

$$\Rightarrow 2\pi f_i = \frac{d\theta_i(t)}{dt}$$

$$\Rightarrow \theta_i(t) = 2\pi \int f_i dt$$

Substitute, f_i value in the above equation.

$$\theta_i(t) = 2\pi \int (f_c + k_f m(t)) dt$$

$$\Rightarrow \theta_i(t) = 2\pi f_c t + 2\pi k_f \int m(t) dt$$

Substitute, $\theta_i(t)$ value in the standard equation of angle modulated wave.

$$s(t) = A_c \cos(2\pi f_c t + 2\pi k_f \int m(t) dt)$$

This is the equation of FM wave.

If the modulating signal is $m(t) = A_m \cos(2\pi f_m t)$, then the equation of FM wave will be

$$s(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

Where :

$$\beta = \text{modulation index} = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$$

$$\int m(t) dt = \int A_m \cos 2\pi f_m t dt = \frac{A_m}{2\pi f_m} \sin 2\pi f_m t \quad \left(\because \int \cos mx dx = \frac{\sin mx}{m} \right)$$

$$\therefore s(t) = A_c \cos \left[2\pi f_c t + 2\pi k_f \times \frac{A_m}{2\pi f_m} \sin(2\pi f_m t) \right]$$

$$= A_c \cos \left[2\pi f_c t + \frac{k_f A_m}{f_m} \sin 2\pi f_m t \right]$$

الانحراف الترددي الأقصى Δf

c) Maximum Frequency deviation $[\Delta f_{max}]$:-

It is the difference between maximum frequency of FM signal to that of Unmodulated Carrier frequency.

It is denoted by Δf_{max} .

أقصى مقدار يتغير به تردد الموجة
الكامل عن قيمته الأصلية f_c

$$\text{i.e., } \boxed{\Delta f_{max} = K_f A_m}$$

Proof: From the definition of Frequency deviation,

$$\Delta f_{max} = \text{Max. Frequency of FM-signal} - \text{Frequency of Carrier signal}$$

$$\Delta f_{max} = f_f(t)_{max} - f_c$$

$$\text{From figure 1, } f_f(t)_{max} = f_c + K_f A_m$$

$$\therefore \Delta f_{max} = f_c + K_f A_m - f_c$$

$$\boxed{\Delta f_{max} = K_f A_m} \text{ Hz} \Leftarrow \text{Indicated in figure 1.}$$

Modulation Index : (β)

It is the ratio of Maximum frequency deviation to that of Frequency of message signal. It is denoted by Symbol ' β '.

$$\text{i.e., Modulation Index in FM} \Rightarrow \boxed{\beta = \frac{\Delta f_{max}}{f_m}} \text{ No units}$$

حاصل نسبة بين الانحراف الترددي الأقصى
و تردد الرسالة المستطوية

(FM) مزايا عيوب Advantages and Disadvantages of Frequency Modulation

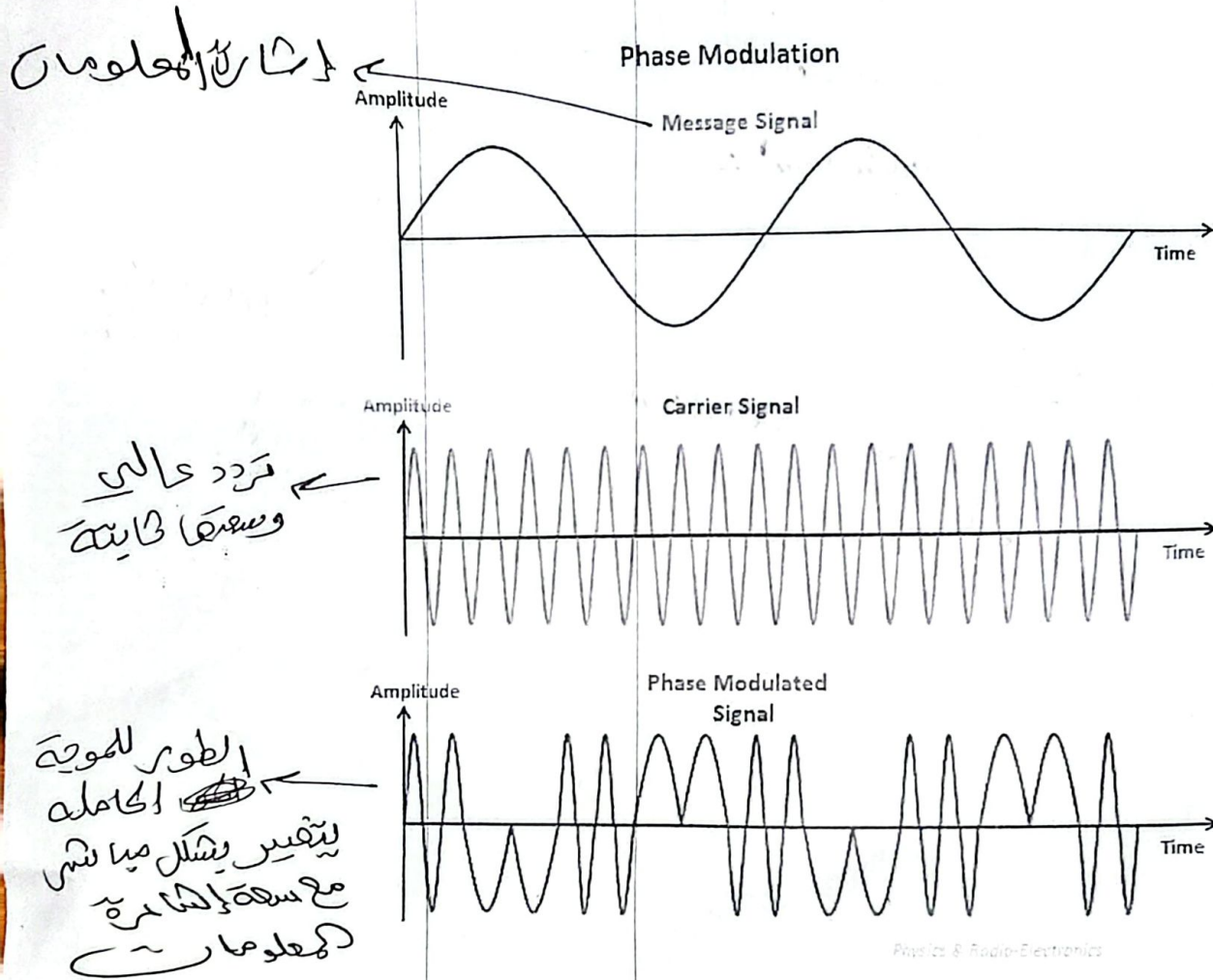
Advantages	Disadvantages
تداخل وتشويش Less interference and noise.	تكلفة أعلى من عرض نطاق ترددي كبير Equipment cost is higher. Has a large bandwidth.
استهلاك طاقة أقل Power consumption is less as compared to AM.	مستقبلات و مرسلات أكثر تعقيداً More complicated receiver and transmitter
القنوات المجاورة معقولة بترددات Adjacent FM channels are separated by guard bands.	يجب أن تكون الهوائيات قريبة لكي ليقل جيداً The antennas for FM systems should be kept close for better communication.

* التشويش يؤثر على جودة الموجات

2. Phase Modulation:

(PM)

The phase of the modulated wave has got infinite points, where the phase shift in a wave can take place. The instantaneous amplitude of the modulating signal changes the phase of the carrier signal. When the amplitude is positive, the phase changes in one direction and if the amplitude is negative, the phase changes in the opposite direction.



In frequency modulation, the frequency of the carrier varies. Whereas, in **Phase Modulation (PM)**, the phase of the carrier signal varies in accordance with the instantaneous amplitude of the modulating signal.

Mathematical Representation

معادلة الطور اللحظي
The equation for instantaneous phase ϕ_i in phase modulation is

$$\phi_i = k_p m(t)$$

التغير في الطور ←

Where,

- k_p is the phase sensitivity
 - $m(t)$ is the message signal
- ← حساسية الطور
← إشارة المعلومات

المعادلة القياسية للموجة المعدلة زاوية
The standard equation of angle modulated wave is

$$s(t) = A_c \cos(2\pi f_c t + \phi_i)$$

Substitute, ϕ_i value in the above equation.

معادلة الموجة المعدلة زاوية
الانحراف الطوري
الاقناعي الذي
تسببه إشارة المعلومات

$$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

← عند معادلة

This is the equation of PM wave.

If the modulating signal, $m(t) = A_m \cos(2\pi f_m t)$, then the equation of PM wave will be

$$s(t) = A_c \cos(2\pi f_c t + \beta \cos(2\pi f_m t))$$

Where,

$$\beta = \text{modulation index} = \Delta\phi = k_p A_m$$

نواير ←

- $\Delta\phi$ is phase deviation
- ↓
β

تقسيم إشارات التردد [FM]

CLASSIFICATION OF FREQUENCY MODULATED SIGNALS:

FM can be divided into ^{نطاق} Narrowband FM and Wideband FM based on the values of modulation index

Narrow band Frequency Modulation (In detail) (NBFM)

→ Narrow band FM signals are characterized by modulation index, β less than 1.

→ Narrow band FM signal equation can be derived from general FM equation for $m(t) = A_m \cos 2\pi f_m t$, as follows

$$S(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)] \quad (1)$$

Equation (1) is general FM equation for $m(t) = A_m \cos(2\pi f_m t)$ obtain for Section 1.2.

قانون بيسيت ← $W.K.T \cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

$$(1) \Rightarrow S(t) = A_c \cos 2\pi f_c t \cos(\beta \sin 2\pi f_m t) - A_c \sin(2\pi f_c t) \times \sin(\beta \sin 2\pi f_m t) \quad (2)$$

(التقريب للزوايا الصغيرة)

For narrow band FM signals, $\beta < 1$

قريب من صفر ← The value of $\beta \sin 2\pi f_m t$ becomes less than 1-degree and it approaches almost 0°. Therefore

$$\cos(\beta \sin 2\pi f_m t) \approx 1 \quad (\because \lim_{\theta \rightarrow 0^\circ} \cos \theta \approx 1) \quad (3)$$

$$\sin(\beta \sin 2\pi f_m t) \approx \beta \sin 2\pi f_m t \quad (\because \lim_{\theta \rightarrow 0^\circ} \sin \theta = \theta) \quad (4)$$

By substituting equations (3) & (4) for equation (2) we get narrow band FM signal

المعادلة ← $S(t) = A_c \cos 2\pi f_c t - A_c \beta \cdot \sin 2\pi f_c t \cdot \sin 2\pi f_m t$

النتيجة ← ∴ Narrow band FM signal consists of 3-frequency components

→ $f_c \Rightarrow$ Carrier signal
 $f_c - f_m \Rightarrow$ Lower side band
 $f_c + f_m \Rightarrow$ Upper side band

} Same as that of standard AM signal.

NBFM

∴ Total transmission Bandwidth of Narrowband FM $\Rightarrow BW_T = 2f_m$

1) Carrier signal $\rightarrow A_c \cos 2\pi f_c t$

2) Upper sideband $\rightarrow f_c + f_m$

3) Lower sideband $\rightarrow f_c - f_m$

عبر
النطاق
التردد

قوله
من موجة
معقدة
والتي هي موجة 3 موجية
بسطة

NBFM

WBFM

Comparison between Narrow band FM and Wide band FM :

Parameters	Narrow Band FM	Wide Band FM
1. Modulation Index β	Less than 1 < 1	Greater than 1 أكثر من واحد ومؤثر
2. Band width B_T	$2f_m$ ↓	$2f_m + 2\Delta f_{max}$ ↑
3. Number of Side bands	Two (2)	Infinity
4. Frequency of Message sig f_m	(مناطقة) 30Hz to 3KHz (النطاق البشري)	30Hz to 15KHz
5. Maximum Frequency Deviation	Δf_{max} 15KHz	مطبات ولبين FM 75KHz (معياري، قليل عال)
6. Application	Speech signal Txn Ex: Mobile Communication	MUSIC signal Txn Ex: FM-stations Channels.

من
النطاققاعدة
كارلسون

هذا النطاق
الواسع هو المميز
الذي ندرجه مقابل
الجودة العالية
للصوت

يسمى NBFM
على أنه
FM =

إشارة كلام
من إشارات صوتية
لافا توكي (Walkie-Talkie)

إشارة موسيقى
من قنوات صوتية FM
لأنها توفر جودة صوتية عالية
بمساحة نطاق أكبر

Problem 1

A sinusoidal modulating waveform of amplitude 5 V and a frequency of 2 KHz is applied to FM generator, which has a frequency sensitivity of 40 Hz/volt. Calculate the frequency deviation, modulation index, and bandwidth.

Solution

Given, the amplitude of modulating signal, $A_m = 5V$

Frequency of modulating signal, $f_m = 2KHz$

Frequency sensitivity, $k_f = 40Hz/volt$

We know the formula for Frequency deviation as

$$\Delta f = k_f A_m$$

Substitute k_f and A_m values in the above formula.

$$\Delta f = 40 \times 5 = 200Hz$$

Therefore, frequency deviation, Δf is 200Hz

The formula for modulation index is

$$\beta = \frac{\Delta f}{f_m}$$

Substitute Δf and f_m values in the above formula.

$$\beta = \frac{200}{2 \times 1000} = 0.1$$

Here, the value of modulation index, β is 0.1 which is less than one. Hence, it is Narrow Band FM.

The formula for Bandwidth of Narrow Band FM is the same as that of AM wave.

$$BW = 2f_m$$

Substitute f_m value in the above formula.

$$BW = 2 \times 2K = 4KHz = 4000 Hz$$

Therefore, the bandwidth of Narrow Band FM wave is 4KHz.

فم - هذا هو نطاق تردد

نطاق تردد
مالية

Characteristics of the FM Frequency Band

- FM operates in the VHF and UHF ranges, which exhibit less noise than the HF and MF ranges used for AM
- Due to the propagation nature of FM signals (**space wave propagation**), the propagation distance is limited to the line-of-sight distance between transmitters and receivers. This makes it possible to operate several independent transmitters on the same frequency without severe interference. This is not the case in AM propagation which depends on successive reflections in the Ionosphere

Advantages of FM

- FM receivers can be equipped with **limiters** to remove amplitude variations caused by noise
- Immunity to additive noise
- The larger the modulation index β , the higher the signal-to-noise ratio

Disadvantages of FM

- Large transmission bandwidth
- Complex transmitters and receivers are needed
- Requires additional hardware to account for the variable effect of noise at different frequencies
- Line-of-sight operation limits the reception/coverage areas

AM

- Amplitude of the carrier is proportional to $m(t)$
- Carrier frequency is constant
- Uniform zero crossings
- Linear process
- Small bandwidth
- Simple implementation
- Less immune to channel noise and interference

FM

- Frequency of the carrier is proportional to $m(t)$
- Carrier amplitude is constant
- Variable zero crossings
- Non-Linear process
- Large bandwidth
- Complex implementation
- Highly immune to channel noise and interference

في الدوائر الكهربائية يمكن حساب القدرة اللحظية $P(t)$ من خلال $V(t)$ والمقاومة R

$$P(t) = V(t)^2 / R$$

$$P(t) = [A_C \cos(2\pi f_m t)]^2 / R$$

أحمد الشاذلي

Analog Communication – Multiplexing

What is Multiplexing?

Multiplexing is a technique that allows the simultaneous transmission of multiple signals through a single channel or link.

Using multiplexing the channel bandwidth can be utilized more efficiently or more signals can be transmitted through a channel at the same time

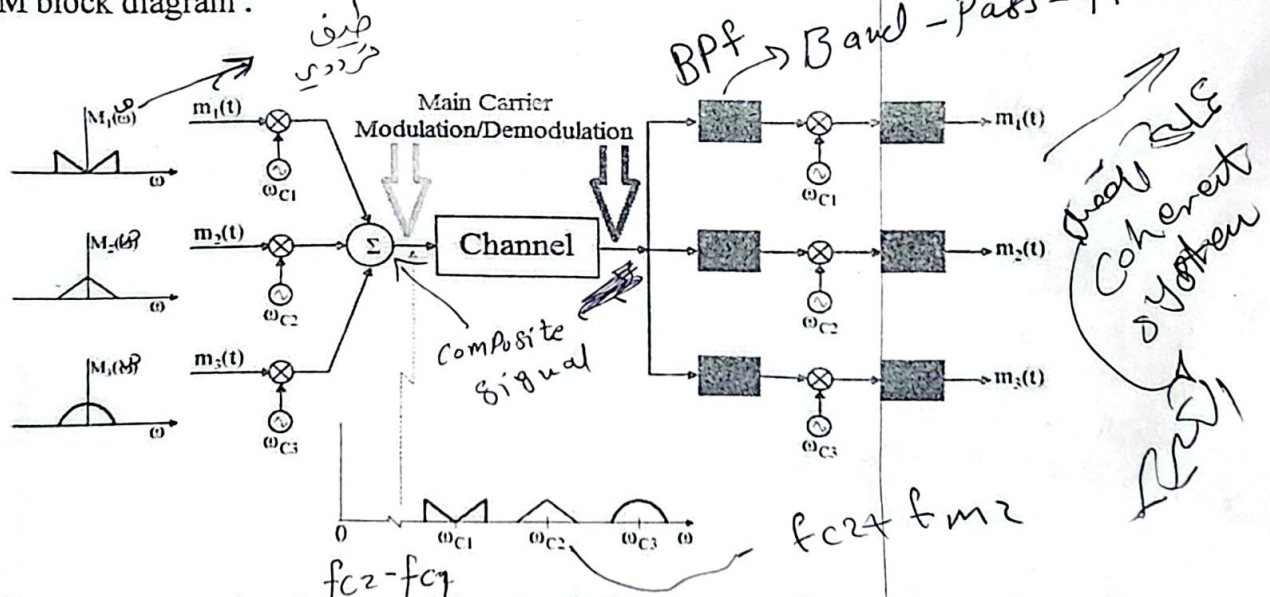
ما هو تقني النقل المتعدد القنوات

What is Frequency Division Multiplexing?

In Frequency Division Multiplexing, the different message signals are modulated at the different carrier frequencies. In this way, the modulated signals are separate from each other in the frequency domain. To avoid the interference between the two message signal, some guard band is also kept between the two message signals.

مخطط كتلة FDM
مخطط كتلة
مخطط كتلة
مخطط كتلة
مخطط كتلة

The FDM block diagram :

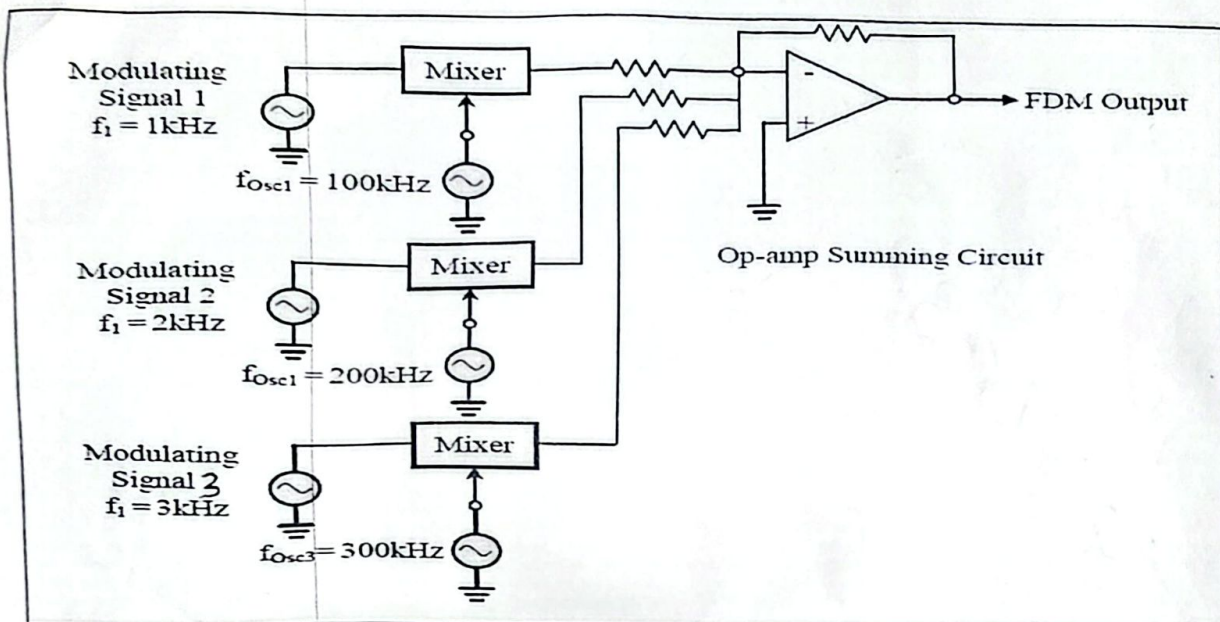


three different message signals are modulated at different carrier frequencies. And then they are combined into a single composite signal. The carrier frequencies of each signal should be selected such that there is no overlapping of two modulated signals. At the receiver, using the bandpass filter, each modulated signal is separated from the composite signal and demultiplexed. And by passing the demultiplexed signal through the low pass filter, it is possible to recover each message signal

Example :

In Frequency Division Multiplexing (FDM) system showing in figure below find the:

1. Bandwidth of each modulating signal
2. Total bandwidth and Guard band
3. Draw the frequency spectrum .

**SIGNAL BW:**

$$BW_1 = 101 - 99 = 2\text{kHz}$$

$$BW_2 = 202 - 198 = 4\text{kHz}$$

$$BW_3 = 303 - 297 = 6\text{kHz}$$

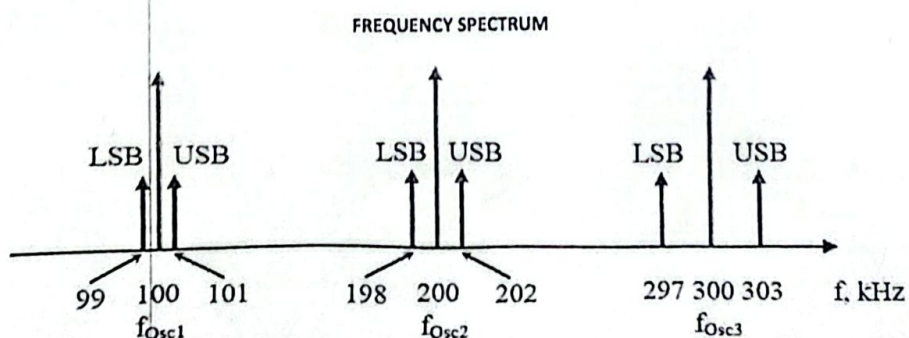
GURD FREQUENCY

$$f_{g1,2} = 198 - 101 = 97\text{kHz}$$

$$f_{g2,3} = 297 - 202 = 95\text{kHz}$$

Total bandwidth

$$BW_T = 303 - 99 = 204\text{kHz}$$

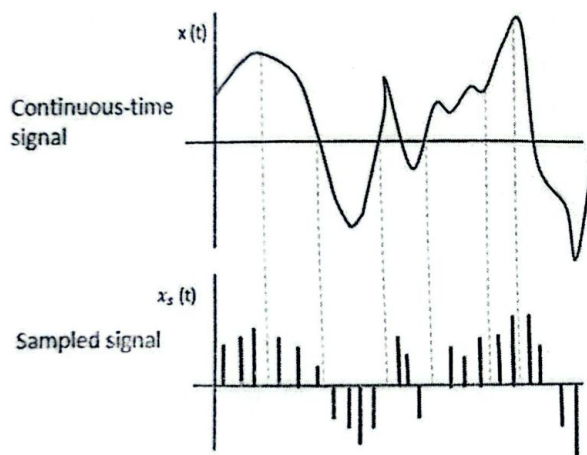


Pulse Analog Modulation.

So far, we have discussed about continuous-wave modulation. Now it's time for discrete signals. The **Pulse modulation** techniques, deals with discrete signals. Let us see how to convert a continuous signal into a discrete one. The process called Sampling helps us with this.

Sampling:

The process of converting continuous time signals into equivalent discrete time signals, can be termed as **Sampling**. A certain instant of data is continually sampled in the sampling process.



The following figure indicates a continuous-time signal $x(t)$ and a sampled signal $x_s(t)$. When $x(t)$ is multiplied by a periodic impulse train, the sampled signal $x_s(t)$ is obtained. A **sampling signal** is a periodic train of pulses, having **unit amplitude**, sampled at equal intervals of time T_s , which is called as the **Sampling time**. This data is transmitted at the time instants T_s and the carrier signal is transmitted at the remaining time.

• Sampling Theorem

While considering the sampling rate, an important point regarding how much the rate has to be, should be considered. The **rate of sampling** should be such that the data in the message signal should neither be lost nor it should get over-lapped.

The **sampling theorem** states that, "a signal can be exactly reproduced if it is sampled at the rate f_s which is greater than or equal to twice the maximum frequency W ."

i.e., Sampling frequency,

$$f_s \geq 2W$$

Where, W = Maximum frequency present in baseband sig.

$$W = \text{Max}(f_{m_1}, f_{m_2}, f_{m_3}, \dots)$$

→ A segment of the signal $x(t)$ is shown in Figure-1(a)

→ Let $S_s(t)$ represents the periodic impulse train and is given by : $S_s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$, shown in Figure 1(b)

Where n , takes all possible values of integers.

$$T_s = \text{Sampling period} = \frac{1}{f_s}$$

→ The reciprocal of Sampling period (T_s) is called sampling rate, $f_s = \frac{1}{T_s}$.

→ The Instantaneous sampling of signal $x(t)$ is indicated in Figure 1(c).

Figure 1(a):
portion of $x(t)$

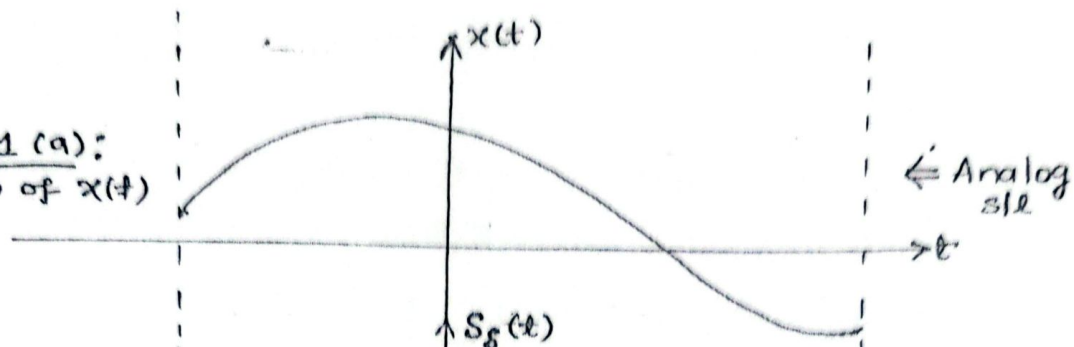


Figure 1(b):
Train of Impulses
 $S_g(t)$

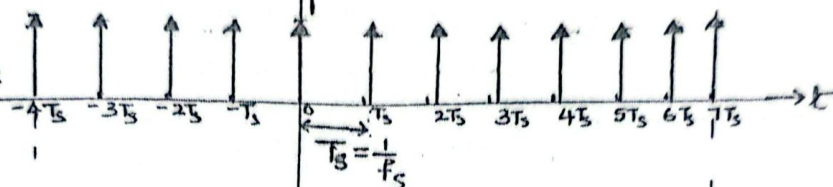
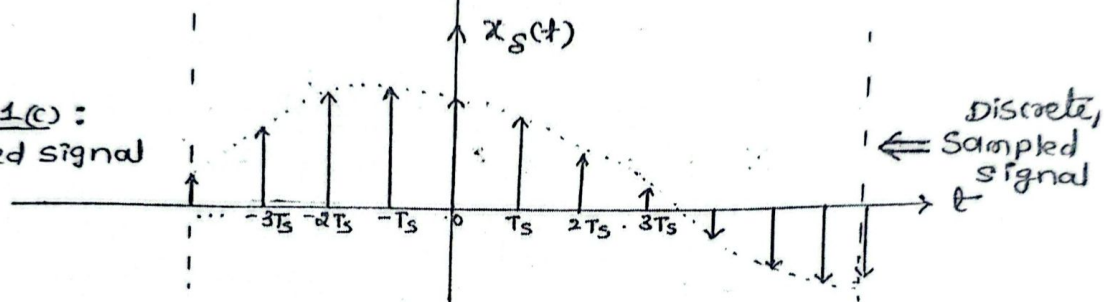


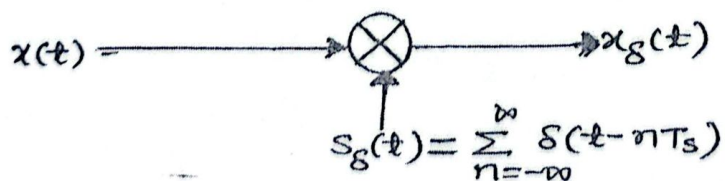
Figure 1(c):
Sampled signal
 $x_g(t)$



Let $x_g(t)$ denotes the signal obtained by multiplying $S_g(t)$ and instantaneous value of $x(t)$ at every T_s seconds.

∴ The sampled signal: $x_g(t)$ is given by,

$$x_g(t) = x(t) \cdot S_g(t) \longrightarrow (1)$$



We may now state the sampling theorem, in two equivalent parts:

1. A Base band signal of finite energy, is completely described by specifying the values of the signal at instant of time separated by $T_s = \frac{1}{f_s} = \frac{1}{2W}$ seconds.
2. A band limited signal of finite energy, may be completely recovered from a knowledge of its samples taken at the rate of " $f_s = 2W$ " samples per second.

NYQUIST RATE:-

A sampling rate of ' $2W$ ' samples per second, for a signal bandwidth of ' W ' Hertz is called Nyquist rate.

i.e., Nyquist Rate = $2W$ $\leftarrow$ It is the minimum sampling frequency required.

NYQUIST INTERVAL:-

The reciprocal of Nyquist rate is called Nyquist Interval. & is measured in seconds.

$$\text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{2W} \text{ seconds}$$

The Concept of UNDERSAMPLING & OVER SAMPLING

- (i) When sampling frequency $f_s = 2W$, then the type of sampling is called critical sampling / Nyquist rate sampling. Results in No-Aliasing effect & No-GUARDBAND

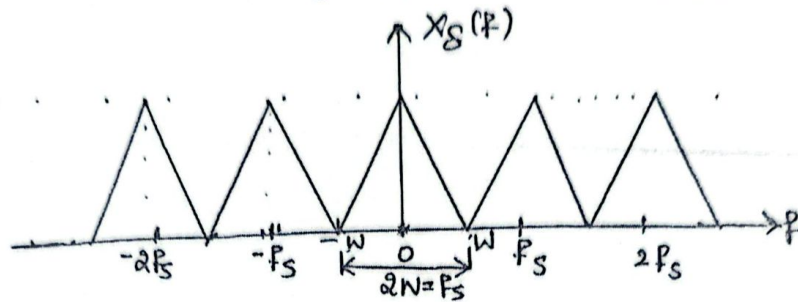


Fig 1: When $f_s = 2W$ [NYQUIST RATE]

- (ii) When $f_s < 2W$:- [UNDERSAMPLING]

When $f_s < 2W$, it results in undersampling & there will be aliasing effect [overlapping]. The spectrum is shown in Fig. 2. In this case it is impossible to recover the signal.

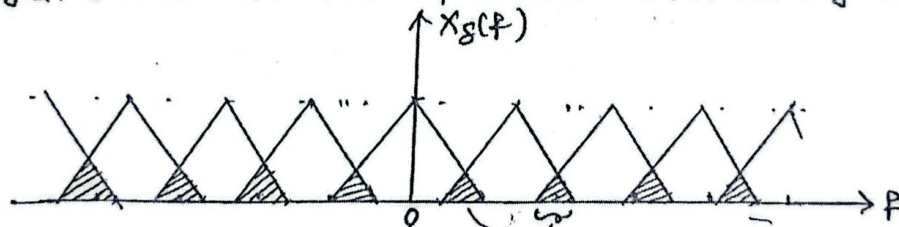
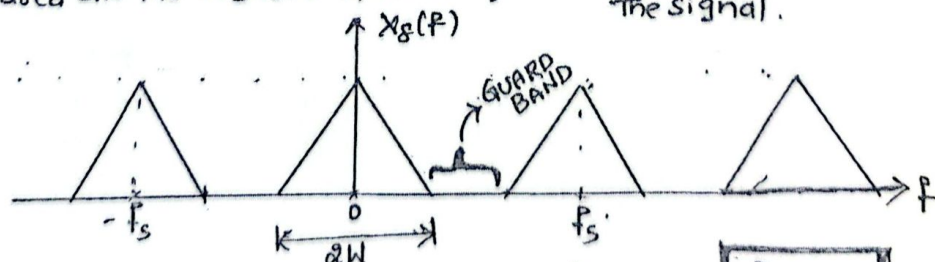


Fig. 2: When $f_s < 2W$

- (iii) When $f_s > 2W$:- [OVER SAMPLING]

When $f_s > 2W$, it results in over sampling & there will be Guard band b/w the adjacent spectrum of $x(t)$. It is easy to recover the signal.



∴ for reconstruction of signal $x(t)$: Sampling frequency $f_s \geq 2W$

Numerical problems

Formulae:

1. Nyquist rate = ' $2W$ ' samples per second @ HZ

where $W = \text{Max}(f_{m1}, f_{m2}, \dots)$ present in given base band signal $x(t)$.

2. Nyquist Interval = $\frac{1}{\text{Nyquist rate}} = \frac{1}{2W}$ seconds.

1. Determine the Nyquist rate and Nyquist interval for: VTU '16/17
June/July
2017

$$(i) \quad m(t) = \frac{1}{\pi t} [\sin(500\pi t)] \quad (ii) \quad g(t) = \text{sinc}(200t)$$

$$\hookrightarrow (i) \quad m(t) = \frac{1}{\pi t} [\sin(500\pi t)]$$

$$\therefore \omega = 500\pi \Rightarrow 2\pi f_m = 500\pi \Rightarrow \boxed{f_m = 250 \text{ HZ} = W}$$

$$\therefore \text{Nyquist rate} = 2W = 2f_m = 500 \text{ HZ} \cdot @ \text{ samples/sec.}$$

$$\therefore \text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{500} \text{ seconds}$$

$$(ii) \quad g(t) = \text{sinc}(200t)$$

$$\text{W.K.T } \text{sinc}(x) = \frac{1}{\pi x} \sin(\pi x)$$

$$\therefore g(t) = \text{sinc}(200t) = \frac{1}{200\pi t} \sin(200\pi t)$$

$$\therefore \omega = 200\pi \therefore 2\pi f_m = 200\pi \Rightarrow f_m = \frac{200}{2} = W = 100 \text{ HZ}$$

$$\therefore \text{Nyquist rate} = 2W = 2f_m = 2 \times 100 = 200 \text{ HZ}$$

$$\text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \frac{1}{200} \text{ seconds}$$

✓

Q2) An analog signal is expressed by the equation.

$x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) + \cos(100\pi t)$. Calculate the Nyquist rate and Nyquist Interval.

→ Given: $x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) + \cos(100\pi t)$

Comparing equation (1) with standard equation → (1)

$$x(t) = 3 \cos(\omega_1 t) + 10 \sin(\omega_2 t) + \cos(\omega_3 t) \rightarrow (2)$$

We get

$$\begin{aligned} \omega_1 &= 50\pi & ; & & \omega_2 &= 300\pi & ; & & \omega_3 &= 100\pi \\ 2\pi f_{m1} &= 50\pi & ; & & 2\pi f_{m2} &= 300\pi & ; & & 2\pi f_{m3} &= 100\pi \\ f_{m1} &= 25 & ; & & f_{m2} &= 150 & ; & & f_{m3} &= 50 \end{aligned}$$

$$\therefore W = \max(f_{m1}, f_{m2}, f_{m3}) = \max(25, 150, 50)$$

$$W = 150 \text{ Hz}$$

$$\therefore \text{Nyquist Rate} = 2W = 2 \times 150 = 300 \text{ Hz @ Samples/sec.}$$

$$\text{Nyquist Interval} = \frac{1}{\text{Nyquist rate}} = \left(\frac{1}{300} \right) = 3.333 \text{ ms}$$

Pulse Amplitude Modulation (PAM)

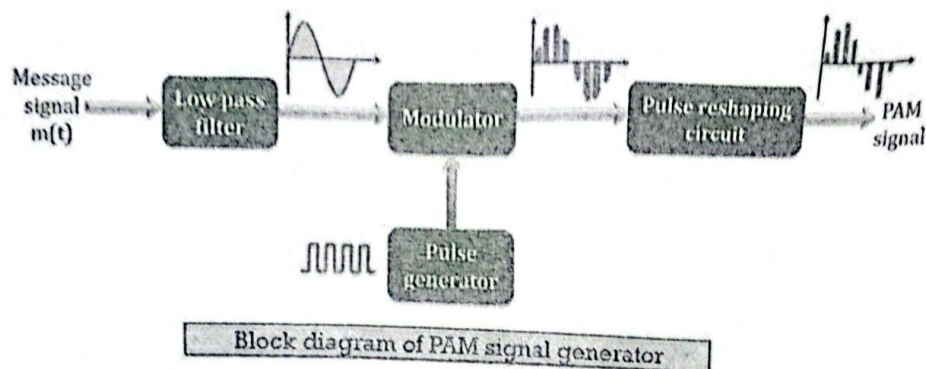
Definition: A modulation technique in which the amplitude of the pulsed carrier signal is changed according to the amplitude of the message signal is known as **Pulse Amplitude Modulation (PAM)**.

Simply put, the transmission of data takes place by the variation of the amplitude of pulse according to the modulating signal.

Block diagram of PAM generator

The figure below shows the block diagram of a PAM generator

As we can see it consist of a low pass filter, a modulator along with a train generator and a pulse reshaping circuit.



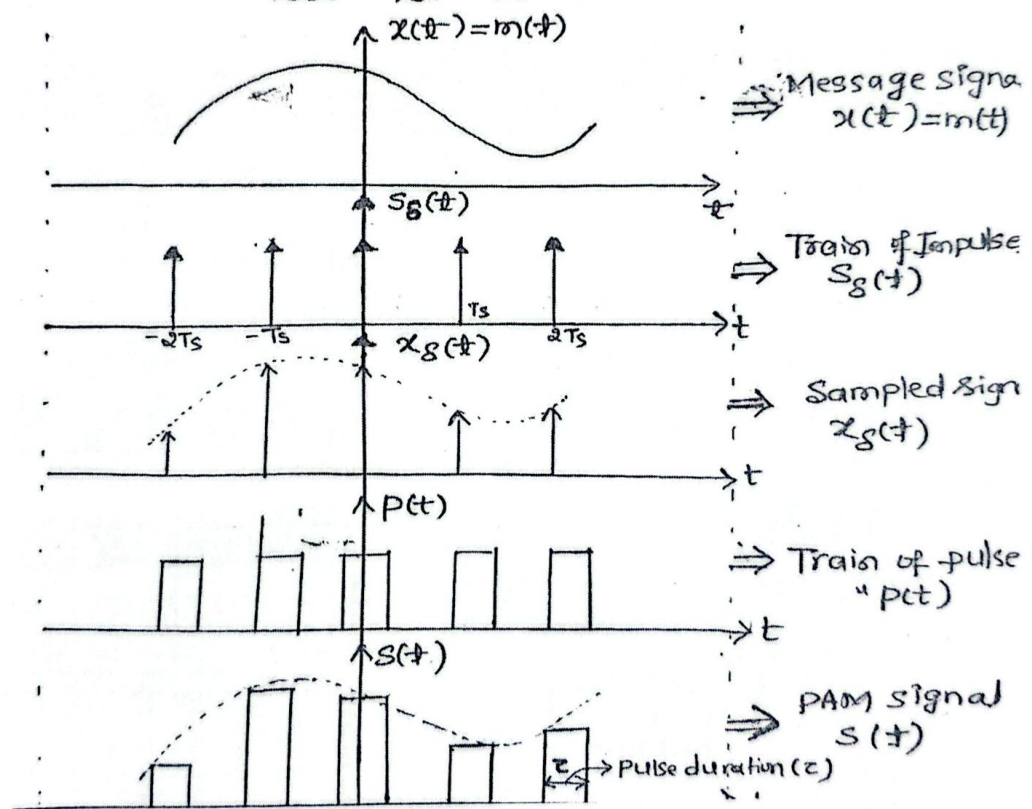
The LPF a Basically, the pulsed carrier gets modulated by the message signal here. The rectangular carrier pulse is generated by the pulse generator circuit. The modulator generates a **pulse amplitude modulated signal**. The sampled pulses can be achieved either by natural or flat top sampling. The output of the modulator is provided to the pulse reshaping circuit. This basically shapes the pulses so that it can be easily detected at the receiver. At the beginning is placed in order to avoid aliasing of the samples

FLAT-TOP SAMPLING:

Q) Explain the generation and recovery of PAM (Flat-Top) signal with necessary equations and spectrum diagram

↳ Definition:- "It is an Analog pulse modulation scheme in which the amplitudes of a train of rectangular carrier pulses are varied in accordance with the sample values of modulating signal."

↳ In PAM, the top of each modulated rectangular pulses is maintained flat. Hence pulse Amplitude Modulation (PAM) is also called as "FLAT TOP SAMPLING"



PAM: Sampling

The instantaneously sampled version of $m(t)$ is given as

$$m_s(t) = \sum_n m(nT_s) \delta(t - nT_s)$$

After some mathematical manipulation, the PAM signal can be expressed as

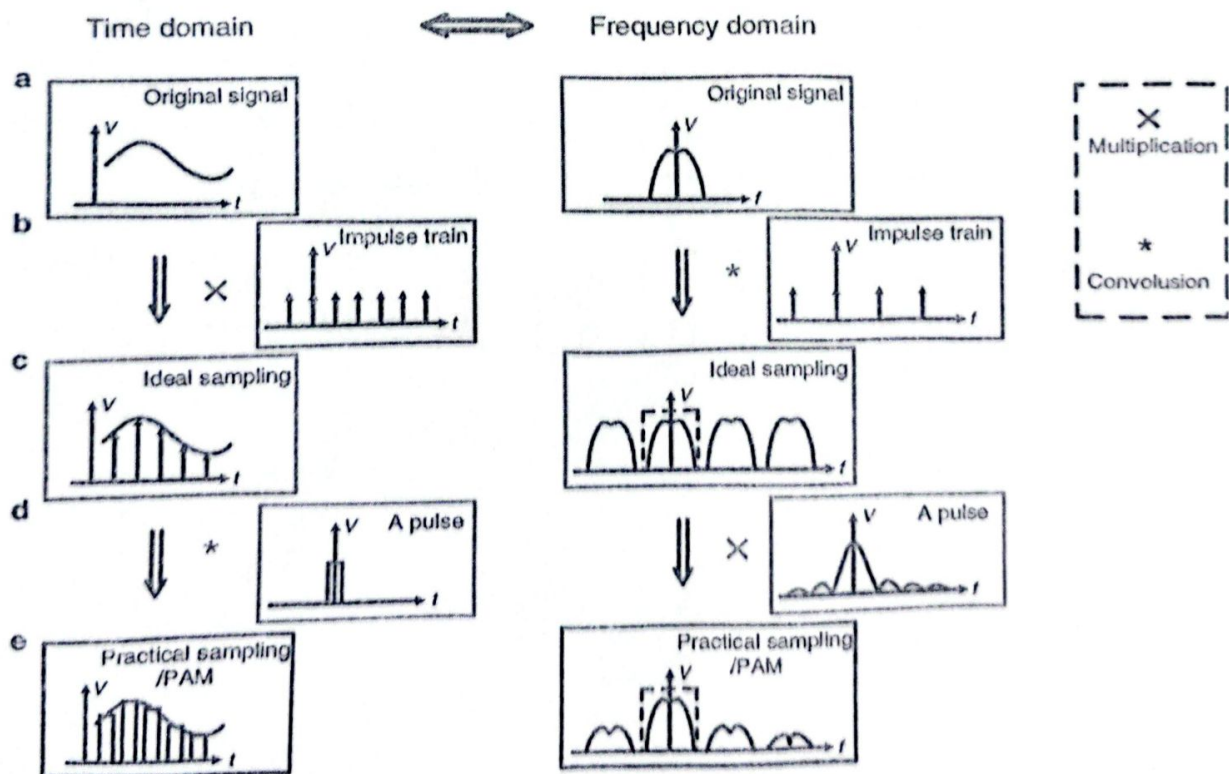
$$s(t) = m_s(t) * h(t)$$

PAM: Flat-Top Rectangular Pulse

For rectangular pulses

$$h(t) = \begin{cases} 1, & 0 < t < T \\ 0, & \text{elsewhere} \end{cases}$$

Representing a time domain and frequency domain of PAM (FLAT TOP), in figure:



Detection of PAM signal :-

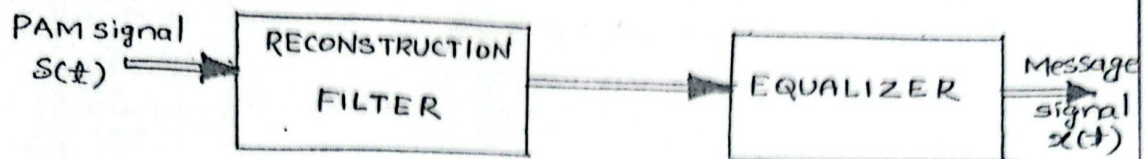


Fig. 2 : Reconstruction of message signal from PAM signal.

The Original message signal, $m(t) = x(t)$ is obtained by passing PAM signal through the reconstruction filter followed by equalizer as shown in Figure. 2.

Message Recovery: LPF

The first step in the recovery of the original signal is through using a **Low-Pass Reconstruction Filter**

The spectrum of the filter output is

$$LPF\{M_s(f).H(f)\} = M(f)LPF\{sinc(fT)e^{-j\pi fT}\}$$

This is an **amplitude distorted** and **$T/2$ delayed** version of the message

Message Recovery: Equalizer

An equalizer is connected in cascade with the LPF and it is used to correct the amplitude distortion

The magnitude response of the equalizer is given as

$$\frac{1}{Tsinc(fT)}$$

$$\text{Duty Cycle} = \frac{T}{T_s}$$

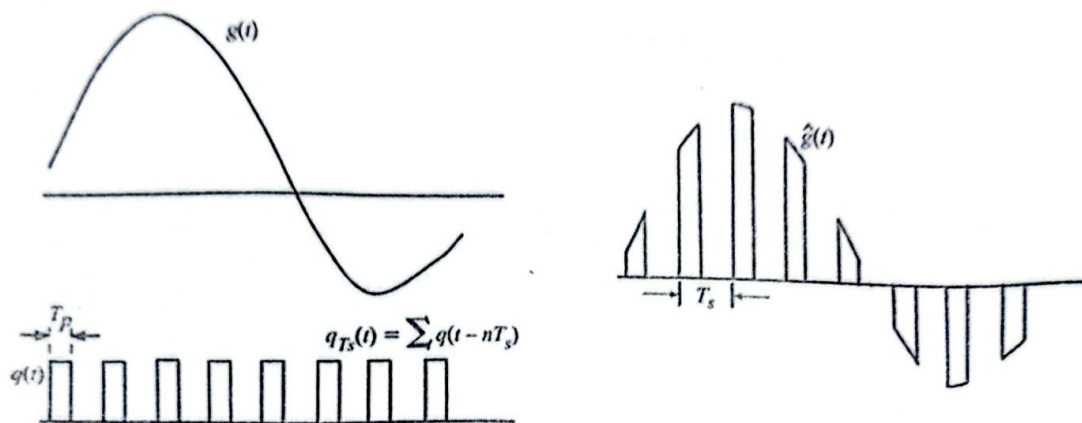
- The presence of T causes the presence of amplitude distortion
- As $T \uparrow$, distortion $\uparrow$
- For duty cycle $\frac{T}{T_s} \leq 0.1$, the amplitude distortion is less than 0.5% and the use of equalization can be ignored.

2. Conventional(natural) PAM

natural sampling is done in case of PAM signals, then the top of the pulses varies according to the modulating signal. So, in this case, the elimination of noise component from the sampled signal becomes somewhat difficult at the time of detection.

At the same time, natural sampling is a bit complex process as compared to flat top sampling.

Hence, flat top sampling is preferred in pulse amplitude modulation.



Advantages of PAM:

- ↳ It is a base for all the digital modulation techniques.
- ↳ Transmitter & Receiver circuitry is simple and easy to construct.

Dis-Advantages of PAM:-

- ↳ It requires high BANDWIDTH for transmission.
- ↳ In PAM signal, amplitude keeps varying, so there is a Noise associated with it.
- ↳ SNR is Less.

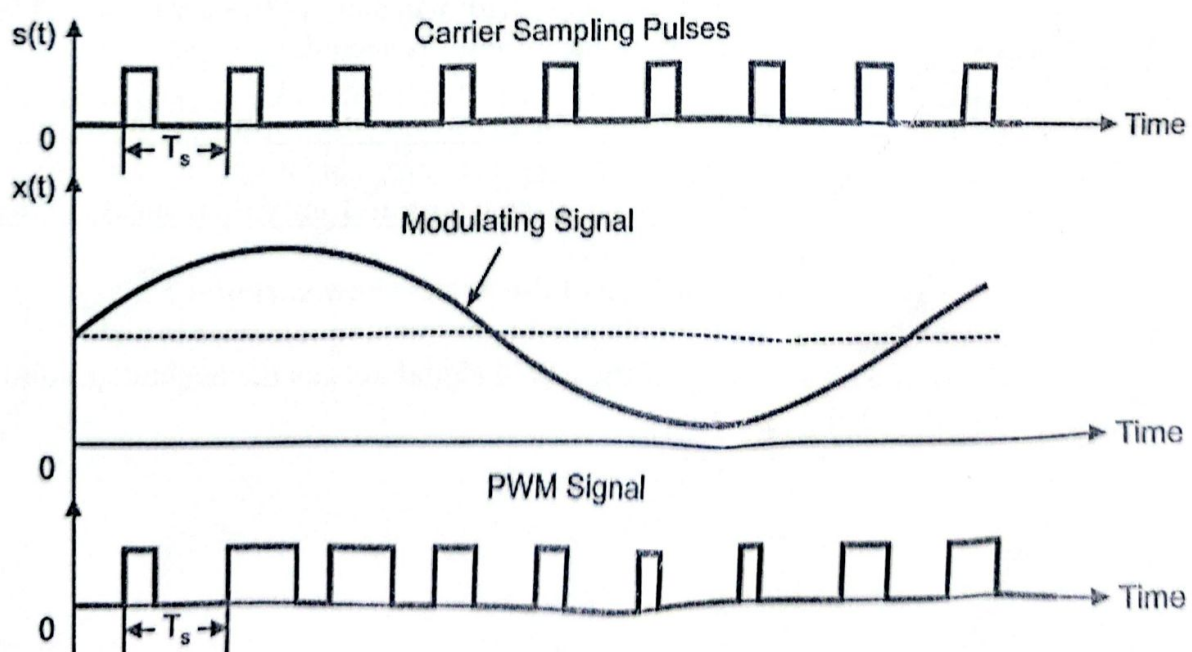
Pulse Width Modulation (PWM)

Definition: A modulation technique where the width of the pulses of the pulsed carrier wave is changed according to the modulating signal is known as **Pulse Width Modulation (PWM)**. It is also known as **Pulse duration modulation (PDM)**.

Basics of Pulse Width Modulation:

It is a type of **Pulse Time Modulation (PTM)** technique where the timing of the carrier pulse is varied according to the modulating signal. In pulse duration modulation (PDM), the amplitude of the pulse is kept constant and only the variation in width is noticed. As the information component is present in width of the pulses. Thus, during signal transmission, the signal undergoes pulse width modulation. Due to constant amplitude property, it gets less affected by noise. However, during transmission channel noise introduces some variation in amplitude as it is additive in nature. But that is totally easy removable at the receiver by making use of limiter circuit.

As the width of the pulses contains information. Thus the noise factor does not cause much signal distortion. Hence the immunity to the noise of a PWM system is better than the PAM system.



Advantages of Pulse Width Modulation

1. It is more immune to channel induced noise than PAM.
2. As noise adds to the amplitude thus the reconstruction of PWM signal from distorted PWM signal is somewhat easy.
3. The transmission and reception do not need to be synchronized.

Disadvantages of Pulse Width Modulation

1. Due to changing width of the pulses, variation in transmission power is also noticed.
2. Bandwidth requirement in case of PWM is somewhat larger than PAM.

Pulse Position Modulation (PPM)

Definition: A modulation technique that allows **variation in the position of the pulses** according to the amplitude of the sampled modulating signal is known as Pulse Position Modulation (PPM). It is another type of PTM, where the amplitude and width of the pulses are kept constant and only the position of the pulses is varied.

Simply put, the **pulse displacement is directly proportional to the sampled value of the message signal.**

Basics of Pulse Position Modulation

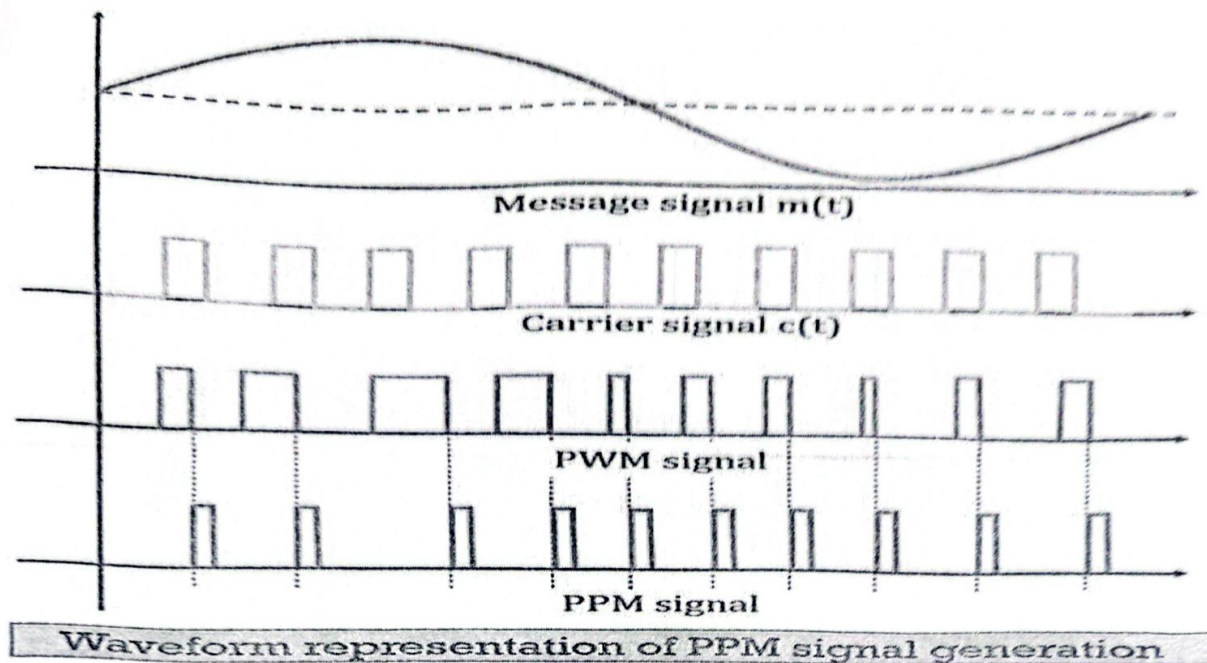
The information is transmitted with the varying position of the pulses in pulse position modulation.

The basic idea about the generation of a PPM waveform is that here, as the amplitude of the message signal increases, the pulse shifts according to the reference.

As we have already discussed in PWM that due to the variable width of the pulses, the transmission power also varies accordingly. However, this is not the case with PPM as here the width of the pulses remains constant and only their position varies. Thus, transmission power does not show variation.

Now, the question arises how the position of the pulses show variation?

As we have already discussed that a PPM signal is generated in reference to a PWM signal. Thus, the trailing edge of the PWM signal acts as the beginning point of the pulses of PPM signal.



Advantages of Pulse Position Modulation

1. PPM shows better noise immunity as compared to PAM & Interference due to noise is more minimal than PAM and PWM, This is so because information content is present in the position of the pulses rather than amplitude.
2. As the amplitude and width of the pulses remain constant. Thus the transmission power also remains constant and does not show variation.
3. Recovering a PPM signal from distorted PPM is quite easy.

Disadvantages of Pulse Position Modulation

1. In order to have proper detection of the signal at the receiver, transmitter and receiver must be in synchronization.

S. No	PAM	PWM/PDM	PPM
1	Amplitude of the pulse is proportional to the amplitude of modulating signal	Width of the pulse is proportional to amplitude of modulating signal.	The relative position of the pulse is proportional to the amplitude of modulating signal.
2	The bandwidth of the transmission channel depends on width of the pulse	Bandwidth of transmission channel depends on rise time of the pulse.	Bandwidth of transmission channel depends on rise time of the pulse.
3	The instantaneous power of the transmitter varies with amplitude of pulses.	The instantaneous power of the transmitter varies with width of pulses	The instantaneous power of the transmitter remains constant with width of pulses.

S. No	PAM	PWM/PDM	PPM
4	Noise interference is high	Noise interference is minimum	Noise interference is minimum
5	System is complex	Simple is implement	Simple is implement
6	Similar to Amplitude modulation	Similar to frequency modulation	Simple to Phase modulation